RE-INSPECTION REPORT

ChArUco target · ceramic · 200 × 150 mm

Re-inspection of CT-IR-000311 (Rev A, 2026-11-04)

ITEM

Part number	CT-CHA-C-200150-6X6
Description	ChArUco calibration target, ceramic, 200 × 150 mm
Serial number	000311
Lot	L-2611-0382
Datasheet	CT-DS-CHA-C · Rev A
Pattern	ChArUco 12 × 17 · 10 / 7 mm DICT_6X6_250
Marker ID range	0–101 (102 markers)
Start ID	0

RE-INSPECTION

Re-inspection date	2027-11-10
Inspector	JG
Location	per Calib.Tools procedure CT-WI-001
Instrument	Optical CMM · OCMM-01
Last calibration	2027-08 · valid to 2028-08
Temperature	20.2 °C ambient · 20 °C reference

OVERALL RESULT

PASS

All measured features within datasheet tolerance.

9 geometry · 1 flatness · 4 visual
176 corners · U = 1.5 µm (k = 2)

1 Method

All 176 ChArUco corners were measured on an optical CMM (OCMM-01); flatness was measured at 25 points. Expanded uncertainty U = 1.5 µm (k = 2). Results are corrected to 20 °C; ambient temperature during measurement 20.2 °C.

2 Pattern geometry

ceramic: deviations and drift in µm

Feature	Nominal (mm)	Previous (mm)	Measured (mm)	Drift (µm)	Deviation (µm)	Tolerance (µm)	Result
Square pitch X	10.000	10.0012	10.0013	+0.1	+1.3	±2	PASS
Square pitch Y	10.000	9.9992	9.9993	+0.1	−0.7	±2	PASS
Pattern width	170.000	170.0040	170.0042	+0.2	+4.2	±5	PASS
Pattern height	120.000	120.0030	120.0028	−0.2	+2.8	±5	PASS
Diagonal 1	208.087	208.0910	208.0911	+0.1	+4.1	±7	PASS
Diagonal 2	208.087	208.0850	208.0849	−0.1	−2.1	±7	PASS
Orthogonality	90.000°	90.0008°	90.0007°	–	–	±0.002°	PASS
Max single-feature deviation	–	–	–	–	1.5	≤ 2	PASS
RMS deviation · information only	–	–	–	–	0.6	–	PASS

3 Flatness

MEASURED FLATNESS (µm)	TOLERANCE (µm)	METHOD	RESULT
13	≤ 20	25 points, 5 × 5 grid, least-squares plane	PASS

4 Visual

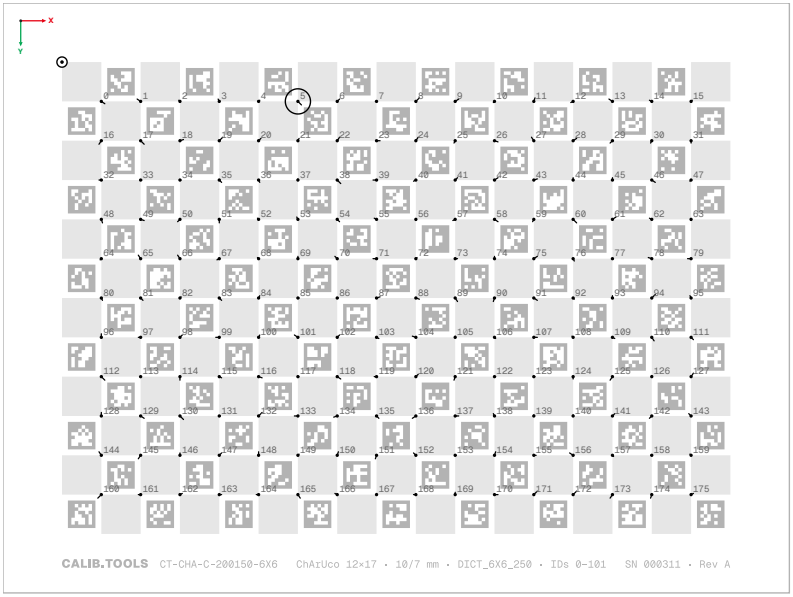
CHECK	CRITERION	RESULT
Edge chamfer	0.5 × 45° on all edges, no chips > 0.3 mm	PASS
Surface defects	No scratches or pinholes in the pattern area	PASS
Print / edge definition	Square edges sharp, no ink bleed at 10× loupe	PASS
Marking legibility	Label line, X / Y arrows and SN legible	PASS

5 Condition on receipt

CHECK	OBSERVATION
Appearance	No visible chips, scratches or contamination observed.
Packaging	Protective foam case intact on receipt.
Customer remarks	[No customer remarks provided]

SAMPLE

6 Feature map



LEGEND

X → right

Y ↓ down

⊙ Origin (0, 0): top-left outer corner of the pattern (OpenCV frame)

• Measured corner with id, n = 176

—— Drift vector, new − previous, magnified × 3000 (1 µm = 3 mm)

◯ Largest drift: id 5, 0.4 µm

7 Measured coordinates

Complete corner coordinates are supplied in CT-CSV-000311-RI01.csv (176 rows; columns id, row, col, x_nom, y_nom, x_meas, y_meas, dx_um, dy_um; mm unless stated). First 10 rows:

id	row	col	x_nom	y_nom	x_meas	y_meas	dx_um	dy_um
0	0	0	10.000	10.000	10.0004	10.0000	+0.4	0.0
1	0	1	20.000	10.000	20.0000	9.9996	0.0	−0.4
2	0	2	30.000	10.000	30.0000	9.9996	0.0	−0.4
3	0	3	40.000	10.000	40.0004	10.0002	+0.4	+0.2
4	0	4	50.000	10.000	50.0001	9.9996	+0.1	−0.4
5	0	5	60.000	10.000	60.0001	9.9997	+0.1	−0.3
6	0	6	70.000	10.000	70.0009	9.9996	+0.9	−0.4
7	0	7	80.000	10.000	79.9997	10.0005	−0.3	+0.5
8	0	8	90.000	10.000	90.0007	9.9994	+0.7	−0.6
9	0	9	100.000	10.000	100.0009	9.9998	+0.9	−0.2

8 Statement

The item identified above meets the specifications of datasheet CT-DS-CHA-C rev A at the time of re-inspection. This is a re-inspection report issued by Calib.Tools; it is not an accredited calibration certificate. An ILAC-MRA certificate issued by an accredited third-party laboratory is available as SVC-X-CERT-ILAC.

9 Traceability

Instrument	OCMM-01 · certificate K-2027-08-0276 · valid to 2028-08
Reference standard	Glass scale GS-01 (200 mm) · certificate K-2027-07-0198 · valid to 2028-07
Raw data	Raw data retained for 10 years under this serial number.

SIGNED

JG

JG · Inspector · 2027-11-10
Printed signature

CALIB.TOOLS SN 000311 INSPECTED 2027-11-10 JG DIMENSIONAL INSPECTION

calib.tools/verify/000311

SAMPLE